

Write a program to implement Logistic Regression (LR) algorithm in python

CODE:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()

X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Logistic Regression Classification")
print("-----")
print("Accuracy:", accuracy)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
```

OUTPUT:

```
*** Logistic Regression Classification
-----
Accuracy: 1.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
```